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universidade de aveiro
departamento de eletrónica,
telecomunicações e informática

Architectural Evolution of nopCommerce

Final Presentation

Scenario C - Omnichannel Commerce Core

TP1 Group 02

Presented by:

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THE PROBLEM

The **nopCommerce** application must stay useful when an external Warehouse Management System (WMS) or Point of Sale System (POS) lag or fail.

Objective is coordination under delay and degradation.

To do this we divided the work in three iterations

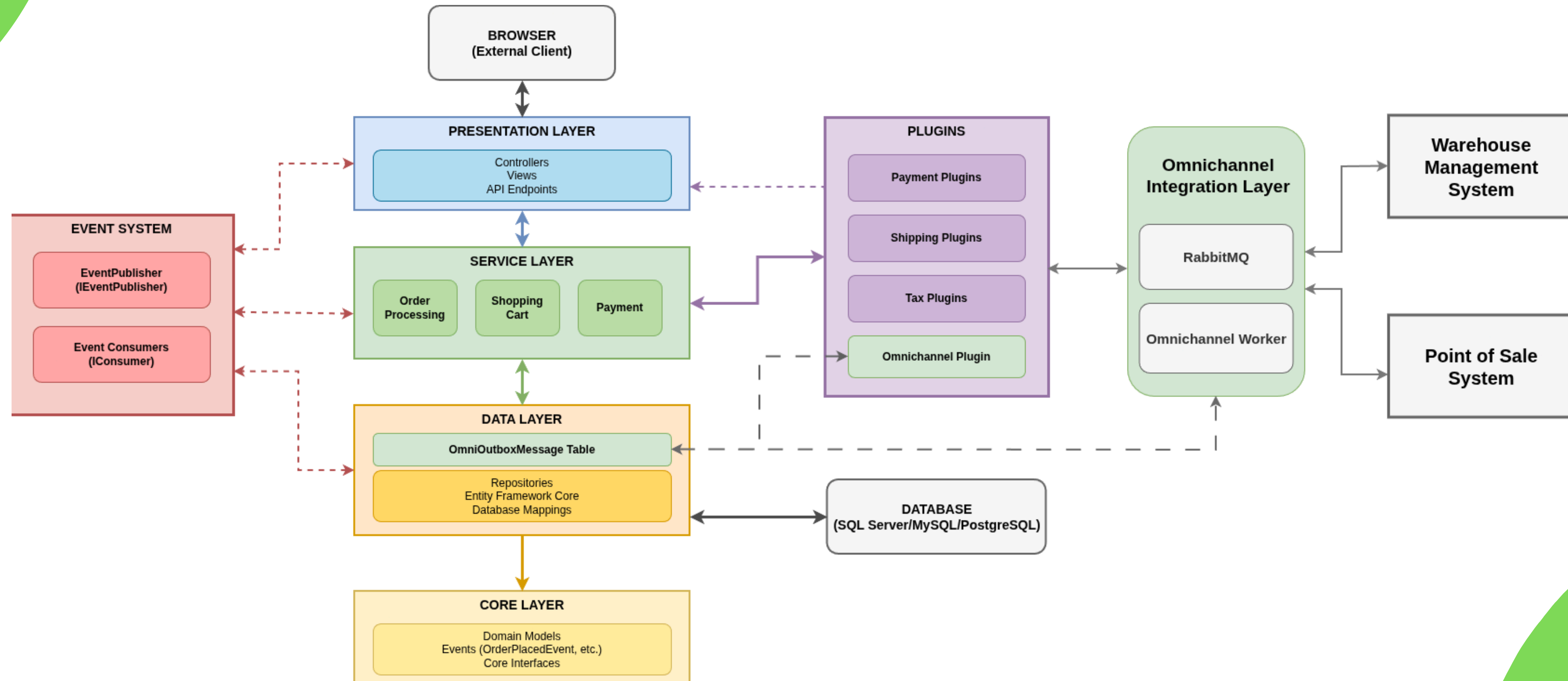
- **Resilience** - checkout still succeeds when the WMS is slow or down
 - **Consistency** - duplicate or out-of-date events never corrupt state
 - **Traceability** - any order is explainable end-to-end from the admin view
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THE APPROACH

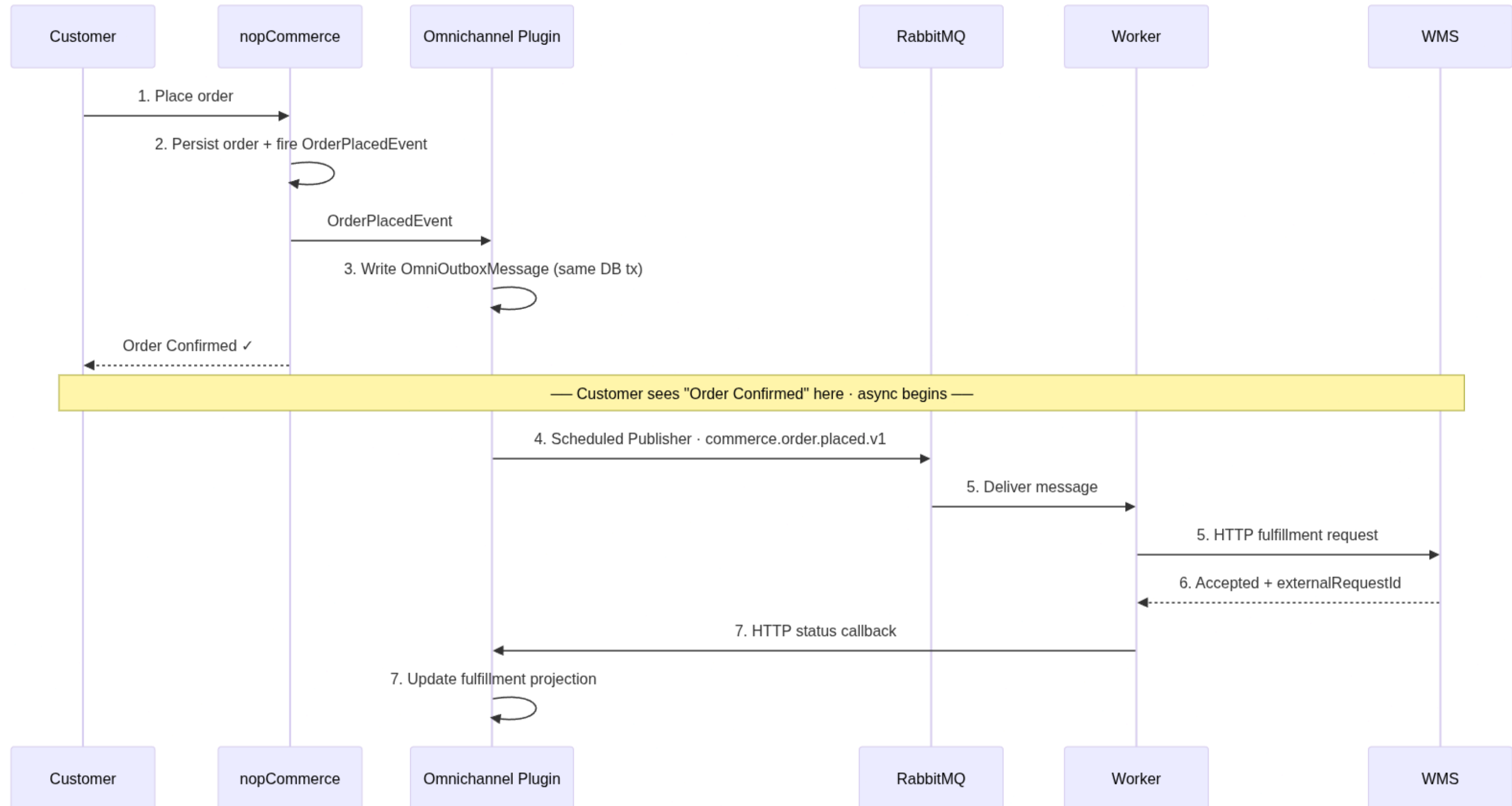
Framework chosen was **Attribute Driven Design (ADD)** with borrowed pieces from ACDM and ADM.

Driver	Description	Iteration
Resilience	checkout survives a WMS outage and self-recovers	1
Performance	integration never blocks the checkout thread	1
Consistency	duplicates or stale events never corrupt state	2
Traceability	any order is explainable end-to-end from admin	3
Opearability	queue, retries & DLQ visible in one dashboard	3

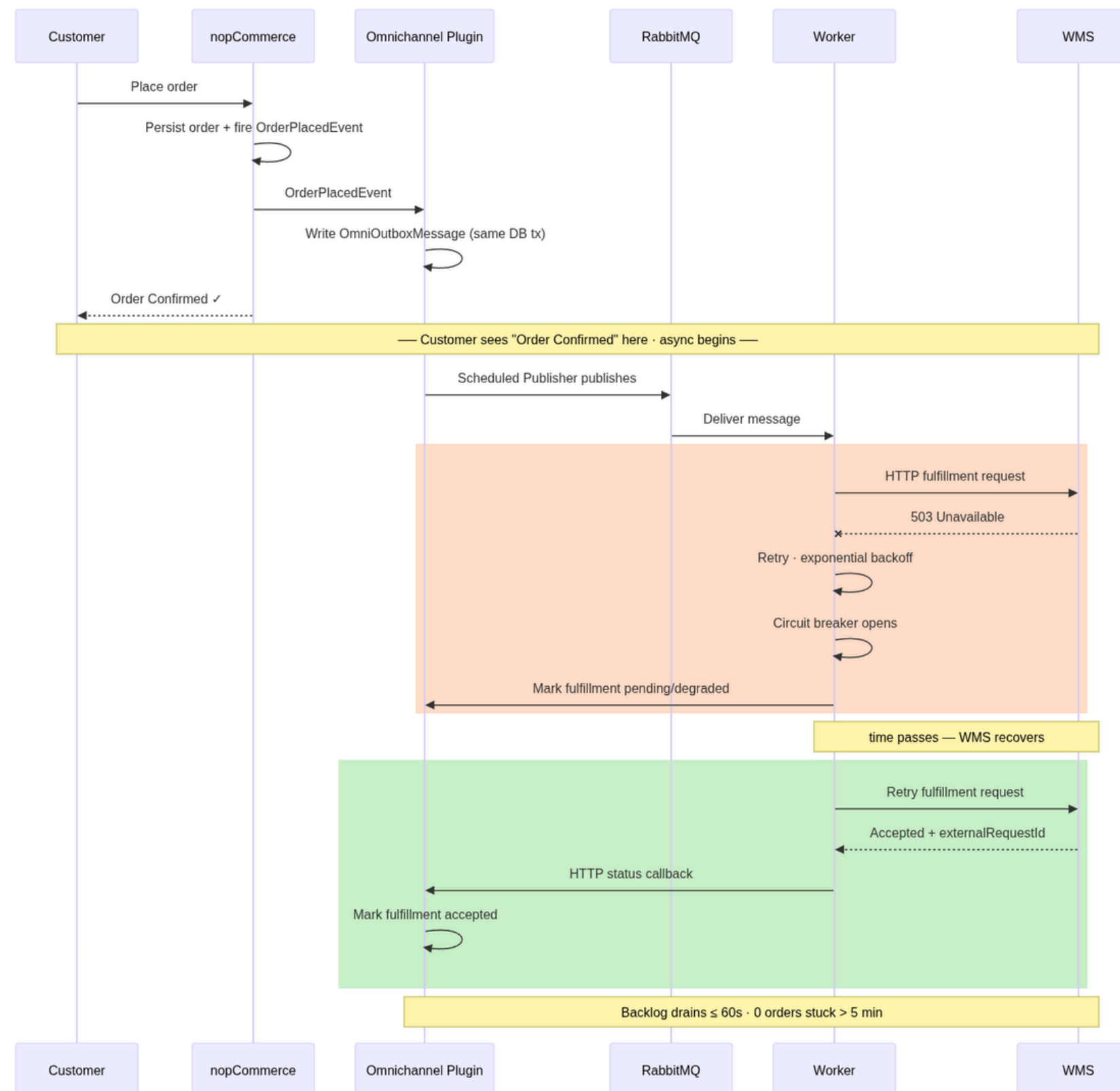
THE ARCHITECTURE

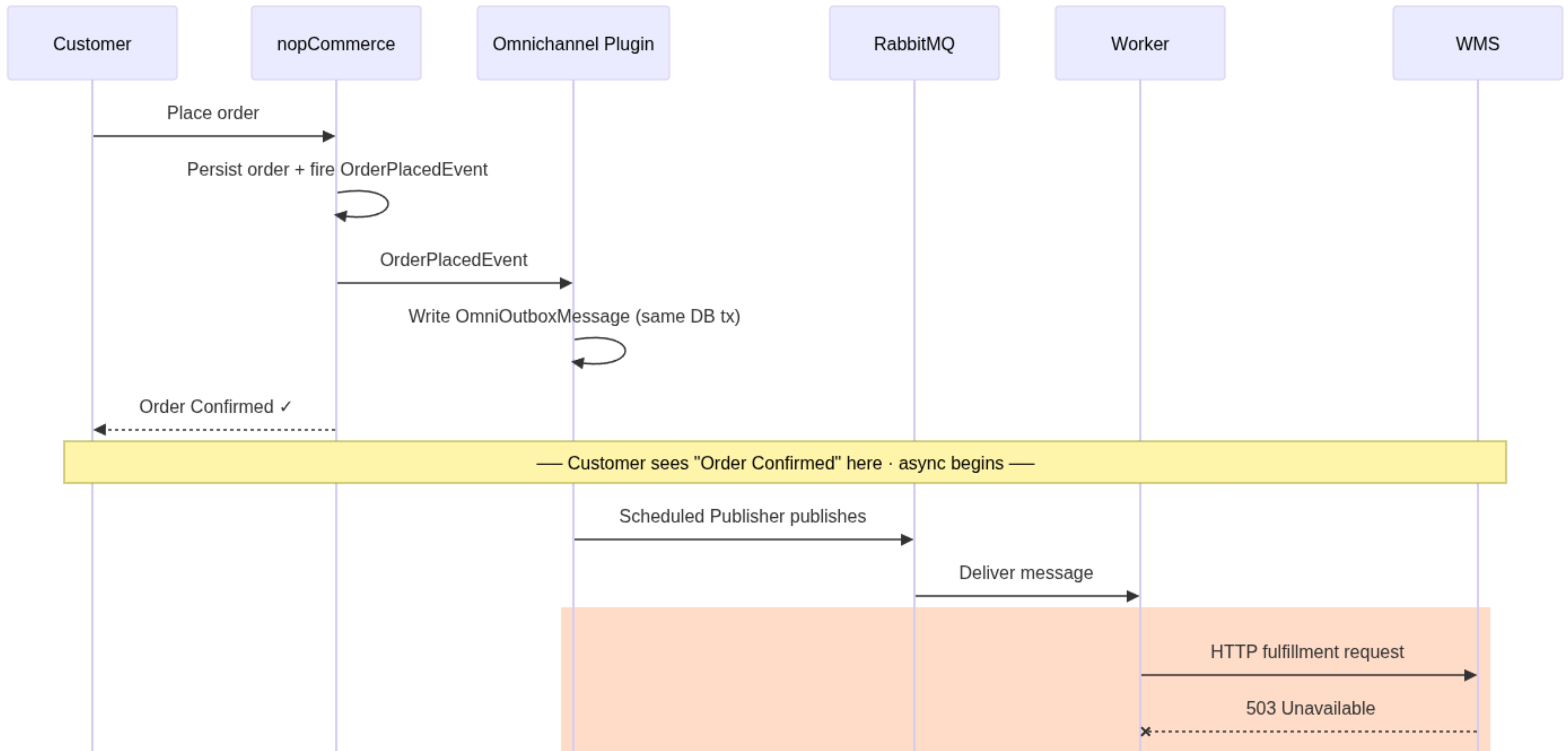


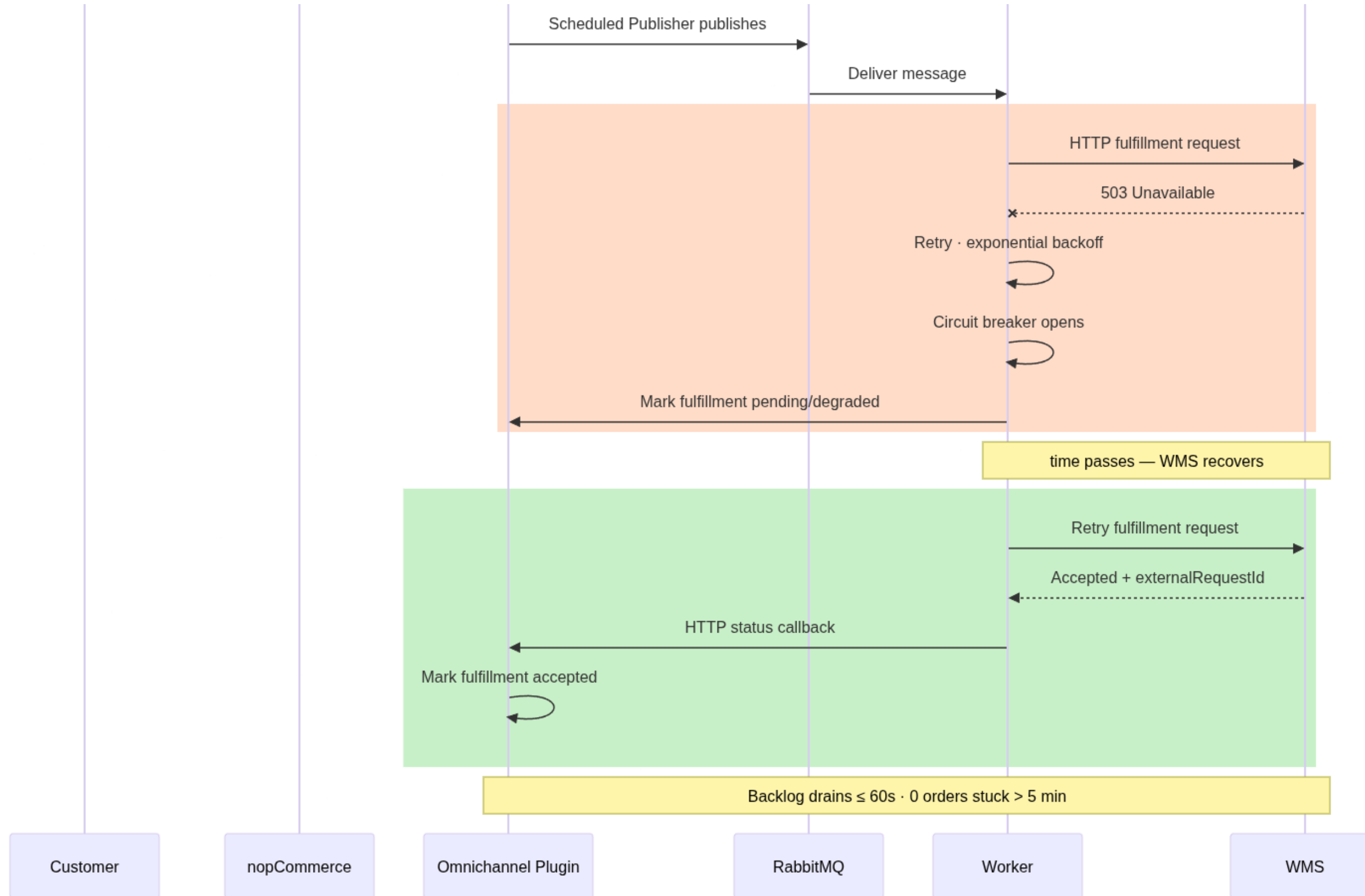
THE ORDER FLOW



THE ORDER FLOW







IMPORTANT DECISIONS

Decision	Tactic	Debt	ADR
Transactional Outbox + Reconsiler	crash-safe event capture	order can't exist with no integration trail	3 + 11
Idempotent Inbox	<i>messageId</i> dedup + <i>sourceVersion</i> guard	at-least-once → no dup , no stale apply	6
Projection-first Stock	shadow table, no write through	stale POS event can't corrupt core stock	7 (partial)
Retry + Circuit Breaker + DLQ	worker → WMS	WMS degration contained, poison isolated	Iter 1



LIVE DEMO



EVIDENCE

	QA	Target	Result
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	Baseline	—	P50 1235 ms · P95 1390 ms ✓
QA-1	resilience	$P95 \leq 2085 \text{ ms}$ under 30 s 503 · drain $\leq 60 \text{ s}$ · 0 pending > 5 min	`TBD`
QA-2	consistency	dup $\leq 50 \text{ ms}$ · 0 dup rows · stale ignored	`TBD`
QA-3	traceability	link by `OrderGuid` · ≤ 3 clicks	`TBD`
QA-4	operability	queue / retry / DLQ / pending in one view · refresh $\leq 5 \text{ s}$	`TBD`
QA-5	performance	outbox $\leq 100 \text{ ms}$ · 0 sync HTTP in checkout	`TBD`

LIMITATIONS

- ****At-least-once**** delivery → mitigated by idempotent inbox ***(not eliminated)***
 - ****Demo-token auth**** on internal callbacks → demo-grade, not production
 - ****No DB unique constraint**** on `messageld` → dedup is application-level
- ****Projection can drift**** from core stock → surfaced as drift, not silently merged
 - ****Inbox / outbox uncapped**** → fine for demo, needs retention in prod
 - ****WMS / POS are simulators**** → contracts versioned for real swap-out

FUTURE WORK

Today (limit)	Next step

Projection can drift from core stock	**Stock write-through** once drift is measured **(ADR-0007)**
App-level `messageld` dedup	**DB unique constraint** on `messageld` **(defense in depth)**
Inbox / outbox uncapped	**Retention + capping** policy
WMS / POS are simulators	**Swap in real systems** via the versioned contracts
Demo-token auth on callbacks	**Real auth** (signed tokens / mTLS)

CHALLENGES

- **`OrderPlacedEvent` fires *after* persistence** → an order could exist with no integration trail → added the **reconciler task** as a safety net **(ADR-0011)**.
- **Publisher-confirms semantics** → publish awaits the broker ack and **throws on nack / timeout**, so the outbox row retries — no silent loss.
- **At-least-once with no DB unique key** → idempotency enforced at the **application layer** (**`messageId` inbox + `sourceVersion` guard**), covered by unit tests.
 - **Breaker thresholds tuned for tests ≠ demo** → thresholds documented and **QA-1 reproduced in demo conditions**, not just CI.
- **.NET 10 + nopCommerce plugin tooling** → **Dockerized build** for a reproducible toolchain across the team.



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